Tyler Winters

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Education

Northeastern University, Boston, MA **Expected Graduation: May 2029**

Candidate for B.S. Mechanical Engineering, ***GPA: 3.91***

- **Relevant Coursework:** Thermodynamics, Fluid Mechanics(in progress), Mechanics of Materials(in progress), Statics, Differential Equations, Engineering Cornerstone

Skills

Software: SolidWorks, AutoCAD, Blender, After Effects, Photoshop
Languages: Python, C, C++
Hardware: Arduino Uno, Raspberry Pi

Projects

Quadruped Walking Robot | 3 Person Team **September 2026 – Present**

- Designed motorized legs in SolidWorks
- Programmed servo motor orientations to move the legs

Automatic Volleyball Dispenser | Personal Project **June 2026 – August 2026**

- Developed a fully automatic ball sensor with a motorized dispenser system using Arduino firmware and PVC piping structure
- Reduced costs of market sources by over **90%**, devising a cheaper alternative for recreational use
- Designed an alternative system using a button to account for light fluctuations

Everglades 3D Hide and Seek Game | Cornerstone Group – Project Website **January 2026 – May 2026**

- Modeled a 3-layer cardboard and papier-mâché replica of the Everglades using AutoCAD
- Presented the exhibit at the Boston Children's Museum to **28** children for them to play
- 3D-printed **22** animals and hid them throughout the structure, where the children competed to find them
- Created realistic rock formations, painted the model, added grass, and designed a water-themed marble run to increase interactivity
- Programmed and wired electronics for a lighting system to illuminate the inside of the structure across Raspberry Pi firmware and LED lighting strips

Force Transport System | Cornerstone Group – Project Website **September 2025 – December 2025**

- Designed a fully automatic force transport system using AutoCAD and laser-cut materials
- Implemented **4** different mechanical components, including a pulley, inclined plane, funnel, and hinge for movement through the system
- Programmed a button and lighting system to sync with the conclusion of our part using Raspberry Pi firmware

Experience

Video Editor **September 2023 – Present**

YouTube | Instagram

- Developed professional-level videos that raised viewership by over **250,000** viewers across various platforms
- Designed thumbnails that increased user interaction by over **60%**
- Implemented Blender 3D modeling and animation to boost viewer retention
- Scripted **300+** short form videos that led to a follower increase of **95%**

Academic Tutor **September 2022 – August 2025**

Hillmont Tutors | Ellicott City, MD

- Designed over **500** math problems for **6** 4th- and 5th-graders to help them jump to advanced-level courses in middle school
- Authored a free study guide for students in similar positions, gaining over **200** users

Volunteering

Animal Care and Organization **September 2023 – December 2025**

Cats on the Grove | Catonsville, MD

- Led management of food produce for a colony of around **20** feral cats
- Created plans, schedules, and stocked inventory for the community